

Cloud Computing Lab — Experiment 1

Installation of Hypervisors and Initiation of VMs with Image File

Field	Details
Name	TODO: Add your name
Roll Number	TODO: Add your roll number
Section	BTECH_CSE_SECB_2023
Course	Cloud Computing Lab
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1. Aim

Install and configure an open-source hypervisor and create/instantiate a Virtual Machine (VM) using an existing image file. Demonstrate the successful installation and execution of the VM.

2. Theory

A **hypervisor** (Virtual Machine Monitor) is software that creates and runs virtual machines by abstracting the physical CPU, memory, storage and network of a host and sharing them among isolated guest operating systems.

- **Type 1 (bare-metal)** hypervisors run directly on hardware (e.g. VMware ESXi, Microsoft Hyper-V, KVM, Xen).
- **Type 2 (hosted)** hypervisors run as an application on top of a host OS (e.g. Oracle VirtualBox, VMware Workstation).

Oracle VirtualBox is a free, open-source (GPLv3) Type 2 hypervisor. It uses hardware virtualization extensions (Intel VT-x / AMD-V) and nested paging to run guest operating systems at near-native speed. On this Windows 11 host, where the Windows Hypervisor Platform (Hyper-V) is active, VirtualBox runs the VM through the Hyper-V backend.

Virtualization is the foundation of cloud computing (IaaS): cloud providers such as AWS EC2, Azure VMs and Google Compute Engine provision VMs from **images** in exactly the same way this lab creates a VM from an ISO image.

3. Requirements

Host system

Item	Value
Host OS	Microsoft Windows 11 Home Single Language, 64-bit
CPU	12th Gen Intel Core i7-12700H (14 cores / 20 threads)
RAM	16 GB (15.63 GB usable)
Free disk space	255 GB on R: drive

Software

Software	Version	Source
Oracle VirtualBox (hypervisor)	7.2.18 r175117	https://www.virtualbox.org/wiki/Downloads
Ubuntu Desktop ISO (guest image)	26.04.1 LTS "Resolute Raccoon", amd64	https://ubuntu.com/download/desktop

ISO integrity was verified against Canonical's official `SHA256SUMS` file:

```
601e30fbf5d97759367c632e2c33630665039b7e2158fd068403da3ccf1bda1f  ubuntu-26.04.1-desktop-amd64.iso
so    (match: OK)
```

4. VM Configuration

Setting	Value
VM name	Ubuntu-Cloud-Lab
Type / Version	Linux / Ubuntu (64-bit)
Firmware	BIOS
CPU	4 virtual CPUs
RAM	6144 MB
Video memory / controller	128 MB / VMSVGA
Storage controller	SATA (AHCI) + IDE (optical)
Virtual hard disk	Ubuntu-Cloud-Lab.vdi, VDI, dynamically allocated, 25 GB
Optical drive	Ubuntu 26.04.1 Desktop ISO (removed after installation)
Boot order	Hard Disk → Optical
Network	Adapter 1: Intel PRO/1000 MT Desktop, NAT (guest IP 10.0.2.15)
Acceleration	Hardware virtualization, Nested Paging, KVM paravirtualization
Guest Additions	7.2.18

The full VirtualBox configuration dump is saved in [vm-configuration.txt](#).

5. Procedure

1. **Install the hypervisor.** Downloaded Oracle VirtualBox 7.2.18 for Windows from the official download page and installed it with default options.
2. **Obtain the image.** Downloaded the Ubuntu 26.04.1 LTS Desktop ISO (Intel/AMD 64-bit) and verified its SHA-256 checksum.
3. **Create the VM.** In VirtualBox Manager → *New*, named the VM `Ubuntu-Cloud-Lab`, Type *Linux*, Version *Ubuntu (64-bit)*.
4. **Configure CPU, RAM, storage and network.** Allocated 4 CPUs and 6144 MB RAM; created a 25 GB dynamically allocated VDI disk on the SATA controller; kept Network Adapter 1 attached to NAT. Equivalent CLI:

```
VBoxManage createvm --name Ubuntu-Cloud-Lab --ostype Ubuntu_64 --register
VBoxManage modifyvm Ubuntu-Cloud-Lab --memory 6144 --cpus 4 --vram 128 --graphicscontroller vmsvga --nic1 nat
VBoxManage createmedium disk --filename Ubuntu-Cloud-Lab.vdi --size 25600 --format VDI
VBoxManage storagectl Ubuntu-Cloud-Lab --name "SATA Controller" --add sata --controller IntelAhci
VBoxManage storageattach Ubuntu-Cloud-Lab --storagectl "SATA Controller" --port 0 --device 0 --type hdd --medium Ubuntu-Cloud-Lab.vdi
```

5. **Attach the ISO.** Attached `ubuntu-26.04.1-desktop-amd64.iso` to the IDE optical drive.

```
VBoxManage storagectl Ubuntu-Cloud-Lab --name "IDE Controller" --add ide
VBoxManage storageattach Ubuntu-Cloud-Lab --storagectl "IDE Controller" --port 0 --device 0 --type dvddrive --medium ubuntu-26.04.1-desktop-amd64.iso
```

6. **Start and boot the VM.** Started the VM; it booted from the ISO into the Ubuntu live environment and the installer.
7. **Install the guest OS.** Installed Ubuntu using VirtualBox's *Unattended Guest OS Install* (the same option offered in the *New VM* wizard), with language English, time zone Asia/Kolkata, hostname `ubuntu-cloud-lab`, user `student`, and the whole virtual disk used for Ubuntu. VirtualBox Guest Additions were installed for better display and host integration.

```
VBoxManage unattended install Ubuntu-Cloud-Lab --iso=ubuntu-26.04.1-desktop-amd64.iso --user=student --full-user-name="Cloud Lab Student" --hostname=ubuntu-cloud-lab.local --locale=en_US --country=IN --time-zone=Asia/Kolkata --install-additions --start-vm=headless
```

8. **Reboot.** After installation the VM rebooted from the virtual hard disk and the ISO was ejected.
9. **Verify the VM.** Logged in to the Ubuntu desktop, opened Terminal and checked the OS, kernel, CPU, memory, disk and network (see Output).
10. **Install build tools and kernel headers** as required by the lab:

```
sudo apt update
sudo apt install -y build-essential linux-headers-$(uname -r)
```

11. **Record configuration and capture screenshots** (Section 7).

6. Output

VM verification

```
$ hostnamectl
  Static hostname: ubuntu-cloud-lab
    Virtualization: oracle
Operating System: Ubuntu 26.04.1 LTS
   Kernel: Linux 7.0.0-31-generic
  Architecture: x86-64

$ lsb_release -a
Distributor ID: Ubuntu
Description:    Ubuntu 26.04.1 LTS
Release:        26.04
Codename:       resolute

$ nproc
4

$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.3Gi        1.3Gi        2.2Gi         49Mi        2.0Gi        3.9Gi

$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2       25G   6.8G   17G   29% /

$ ip -brief -4 addr
lo                UNKNOWN        127.0.0.1/8
enp0s3            UP             10.0.2.15/24

$ ping -c 2 ubuntu.com
64 bytes from website-content-cache-2.ps5.canonical.com (185.125.190.21): icmp_seq=2 ttl=64
```

Build tools and kernel headers

```
$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease

$ sudo apt install -y build-essential linux-headers-$(uname -r)
build-essential is already the newest version (12.12ubuntu2.26.04.2).
linux-headers-7.0.0-31-generic is already the newest version (7.0.0-31.31).

$ gcc --version | head -1
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
$ make --version | head -1
GNU Make 4.4.1
$ ls -ld /usr/src/linux-headers-$(uname -r)
/usr/src/linux-headers-7.0.0-31-generic
```

7. Screenshots



Figure 1: Oracle VirtualBox installed and opened

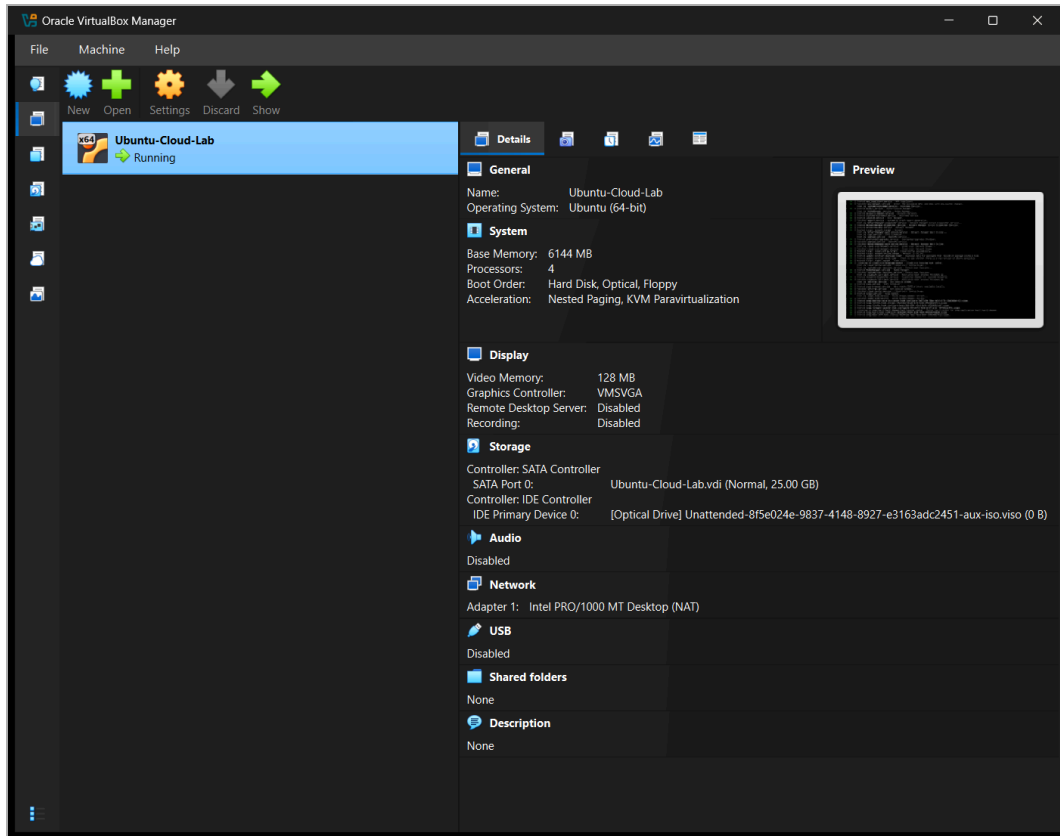


Figure 2: VM configuration: CPU, RAM, storage, network, install media attached

```
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
Starting plymouth-quit-wait.service - Hold until boot process finishes up...
[ OK ] Started networkd-dispatcher.service - Dispatcher daemon for systemd-networkd.
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
Starting setvtrng.service - Set console scheme...
[ OK ] Started cups.service - CUPS Scheduler.
[ OK ] Started cups-browsed.service - Make remote CUPS printers available locally.
[ OK ] Finished setvtrng.service - Set console scheme.
[ OK ] Finished cloud-config.service - Cloud-init: Config Stage.
[ OK ] Started snapd.service - Snap Daemon.
Starting snapd.hold.service - Holds Snappy daemon refresh...
[ OK ] Finished snapd.hold.service - Holds Snappy daemon refresh.
[ OK ] Started snap.desktop-security-center.hook.config-7a871a74-78ea-4e5f-b778-18bb3ddaf413.scope.
[ OK ] Started snap.firefox.hook.install-05e2c0e2-01a6-4b70-bc81-05b3463227c4.scope.
[ OK ] Started snap.firefox.hook.config-beacc3b6-b54f-43cd-8a7a-e32a968c1eb7.scope.
[ OK ] Started snap.firmware-updater.hook.config-2571d7f3-9d48-474f-ac41-709903cac97b.scope.
[ OK ] Listening on snap.hwctl.hwctl-daemon.hwctl-socket.socket - Socket hwctl-socket for snap application hwctl.hwctl-daemon.
[ OK ] Started snap.hwctl.hook.config-0bd4ad91-f8d7-4bd8-9545-9fb5e97fd828.scope.
[ OK ] Started snap.mesa-2404.hook.install-4a200136-7dc5-4ecd-85df-36039d83751d.scope.
[ OK ] Started snap.prompting-client.hook.config-82c678fa-825d-4c7d-9254-9475078b5397.scope.
[ OK ] Started snap.snap-store.hook.config-3c1d0575-25e1-4ab4-9c3d-1f0447aaab0c.scope.
[ OK ] Started snap.snapd-desktop-integration.hook.config-695de251-9f6a-44b8-b1ce-47e5f4eabdf.scope.
[ OK ] Started snap.thunderbird.hook.install-08d91295-3add-4f4d-846e-2f46f24078ed.scope.
[ OK ] Started snap.thunderbird.hook.config-172cb475-832b-4af6-96a6-8255f290a7a1.scope.
[ OK ] Started snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server.
[ OK ] Started snap.ubuntu-desktop-bootstrap.hook.config-e29e153c-1373-45b4-bcec-0109710d0012.scope.
Stopping snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server...
[ OK ] Stopped snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server.
[ OK ] Started snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server.
Starting systemd-timedated.service - Time & Date Service...
[ OK ] Started systemd-timedated.service - Time & Date Service.
[ OK ] Finished snapd.seeded.service - Wait until snapd is fully seeded.
Starting gdm.service - GNOME Display Manager...
[ OK ] Reached target multi-user.target - Multi-User System.
Starting cloud-final.service - Cloud-init: Final Stage...
Starting power-profiles-daemon.service - Power Profiles daemon...
Starting casper-md5check.service - casper-md5check Verify Live ISO checksums...
[ OK ] Finished cloud-final.service - Cloud-init: Final Stage.
[ OK ] Reached target cloud-init.target - Cloud-init target.
[ OK ] Started power-profiles-daemon.service - Power Profiles daemon.
[ OK ] Started gdm.service - GNOME Display Manager.
[ OK ] Reached target graphical.target - Graphical Interface.
[ OK ] Created slice user-1000.slice - User Slice of UID 1000.
Starting user-runtime-dir@1000.service - User Runtime Directory /run/user/1000...
[ OK ] Finished user-runtime-dir@1000.service - User Runtime Directory /run/user/1000.
Starting user@1000.service - User Manager for UID 1000...
[ OK ] Started user@1000.service - User Manager for UID 1000.
[ OK ] Started session-1.scope - Session 1 of User ubuntu.
Starting rtkit-daemon.service - RealtimeKit Scheduling Policy Service...
```

Figure 3: VM booting from the Ubuntu ISO (live environment)

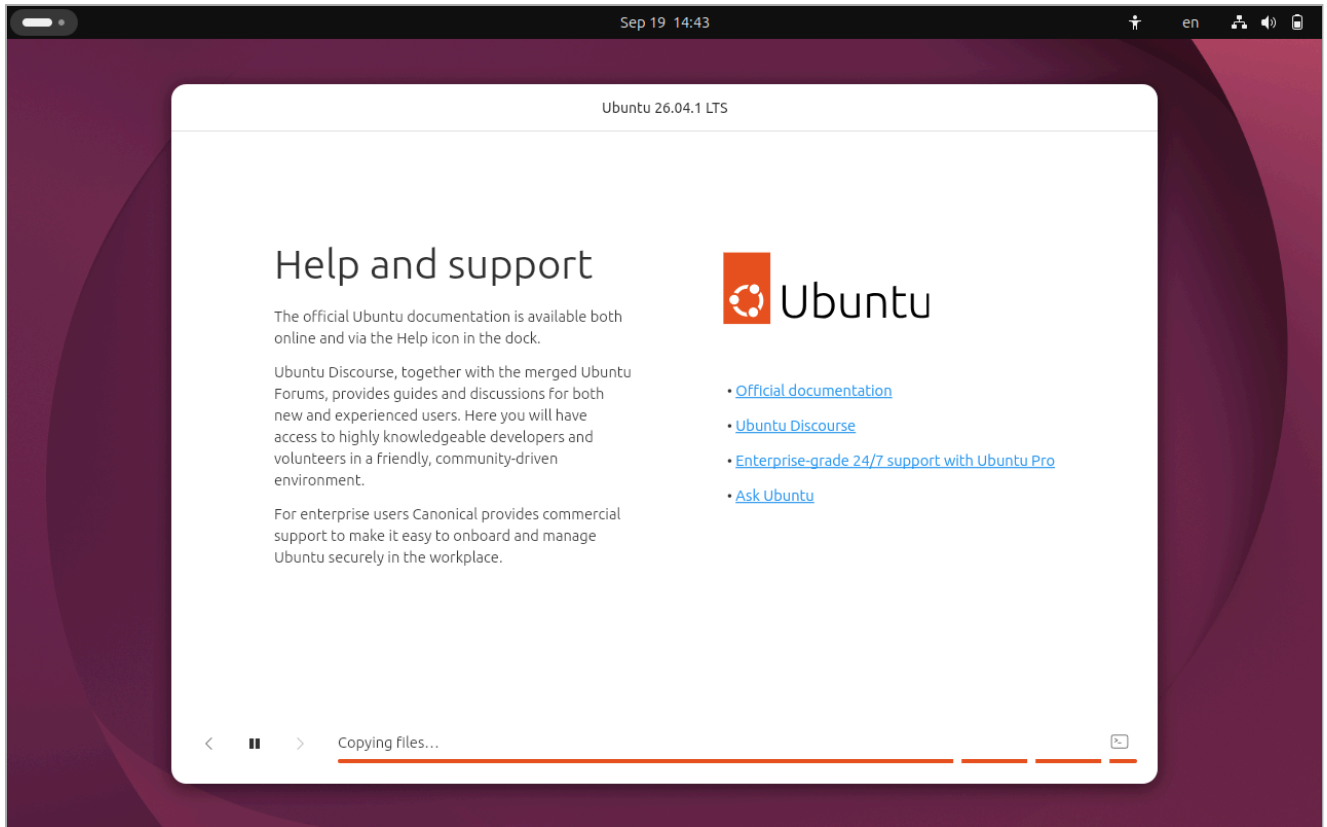


Figure 4: Ubuntu 26.04.1 LTS installer copying files

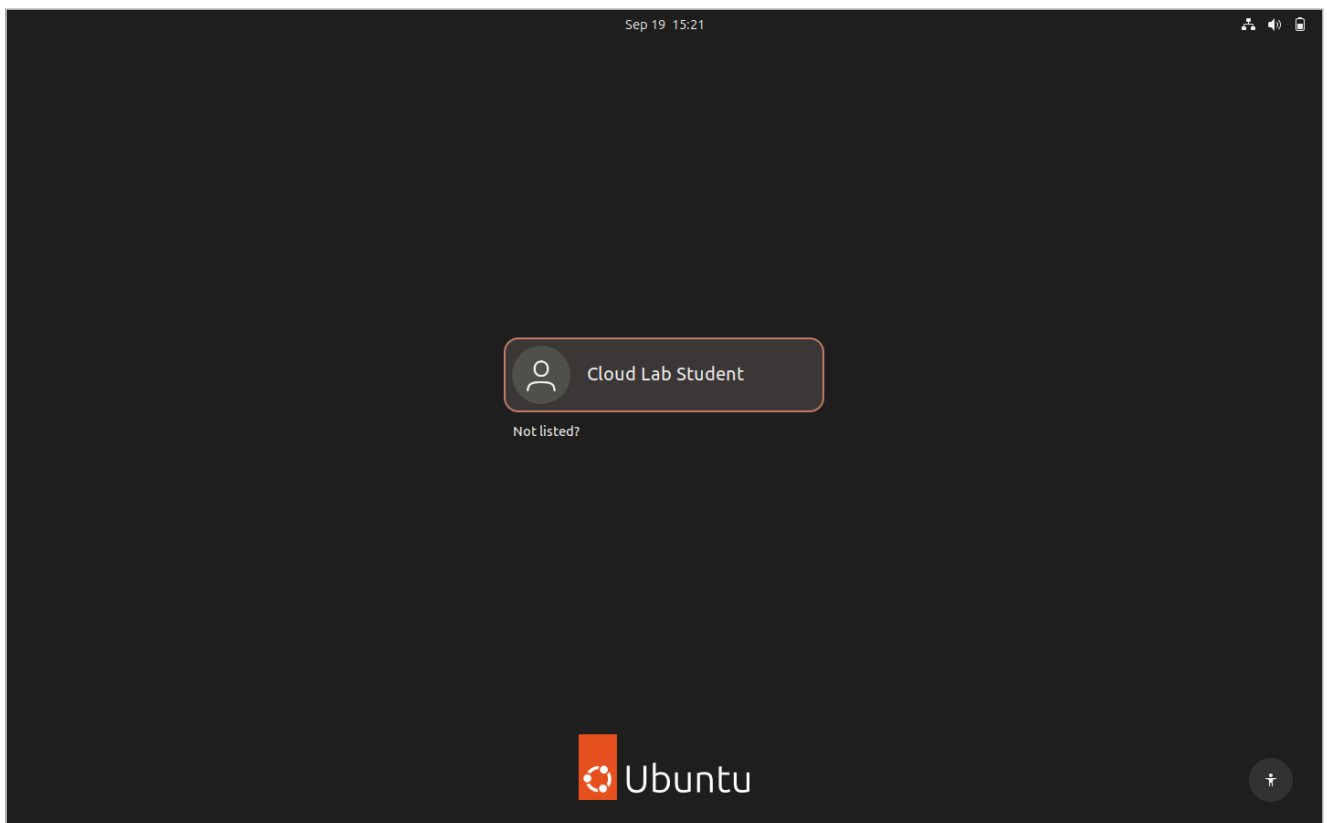


Figure 5: Installed Ubuntu login screen

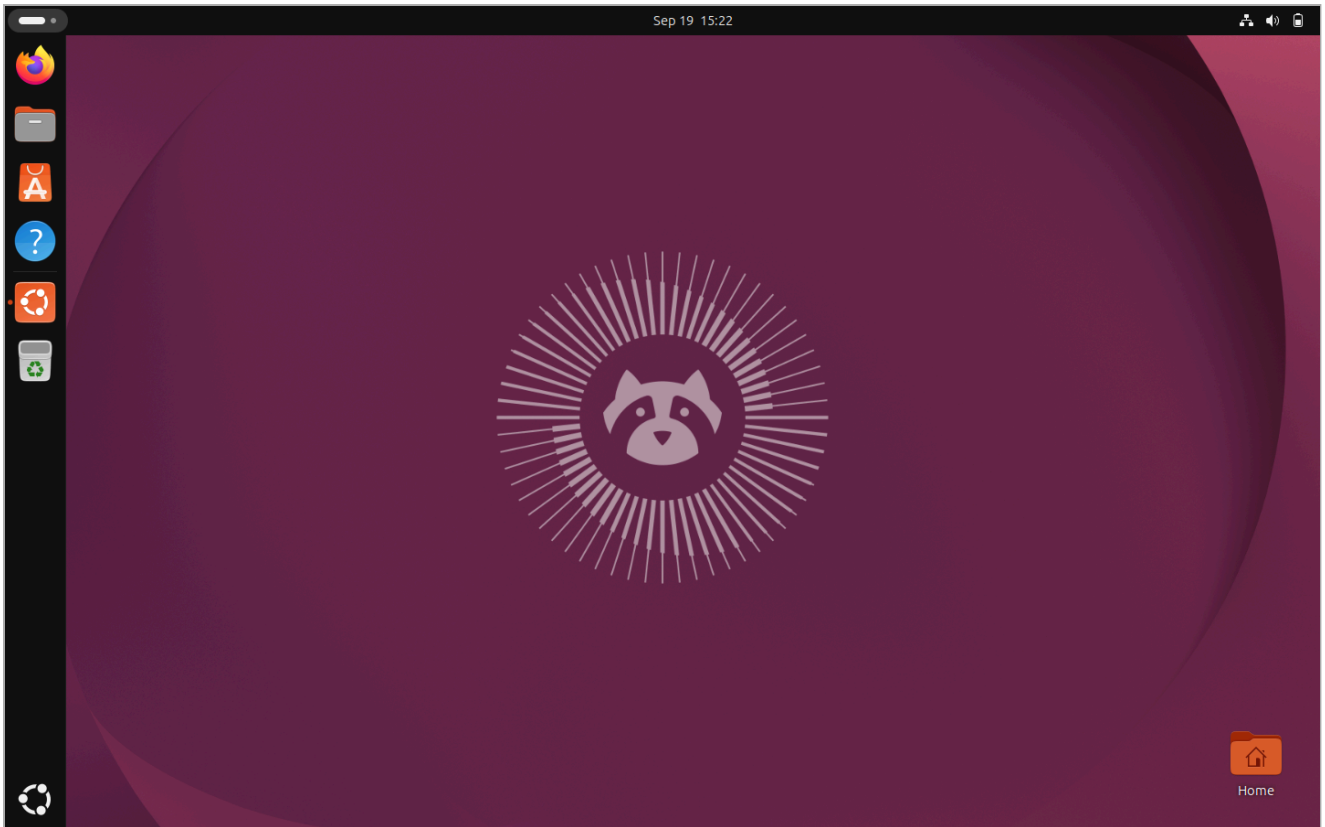


Figure 6: Ubuntu desktop running inside the VM

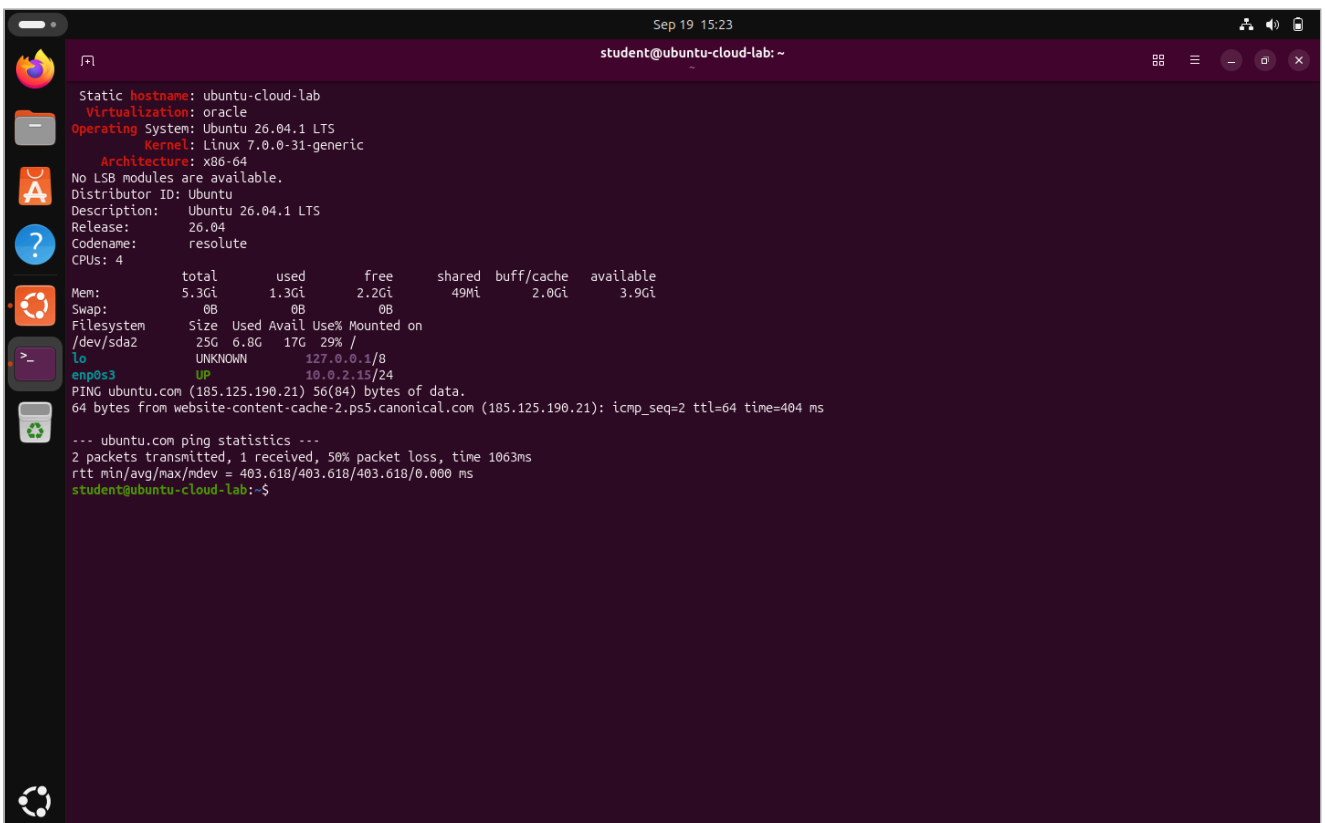


Figure 7: VM verification in Terminal

```
student@ubuntu-cloud-lab: ~  
student@ubuntu-cloud-lab:~$ sudo apt update  
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease  
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease  
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease  
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease  
37 packages can be upgraded. Run 'apt list --upgradable' to see them.  
student@ubuntu-cloud-lab:~$ sudo apt install -y build-essential linux-headers-$(uname -r)  
build-essential is already the newest version (12.12ubuntu2.26.04.2).  
linux-headers-7.0.0-31-generic is already the newest version (7.0.0-31.31).  
linux-headers-7.0.0-31-generic set to manually installed.  
The following packages were automatically installed and are no longer required:  
  linux-image-unsigned-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30  linux-tools-7.0.0-30-generic  
Use 'sudo apt autoremove' to remove them.  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 37  
student@ubuntu-cloud-lab:~$
```

Figure 8: `sudo apt update` and `sudo apt install -y build-essential linux-headers-$(uname -r)`

```
student@ubuntu-cloud-lab: ~  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
g++ (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
GNU Make 4.4.1  
ii build-essential 12.12ubuntu2.26.04.2 amd64      Informational list of build-essential packages  
ii linux-headers-7.0.0-31-generic 7.0.0-31.31      amd64      Linux kernel headers for version 7.0.0  
/usr/src/linux-headers-7.0.0-31-generic  
student@ubuntu-cloud-lab:~$
```

Figure 9: `gcc`, `make` and kernel headers verified

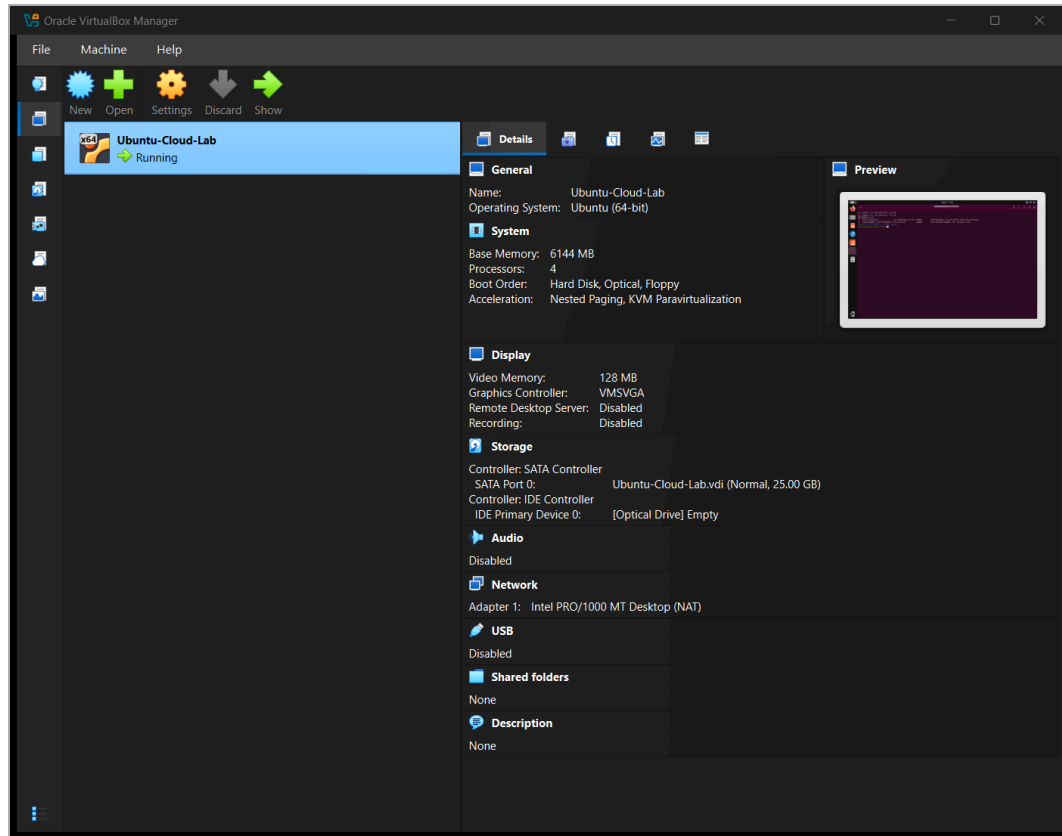


Figure 10: VM running after installation (ISO ejected)

8. Result

Oracle VirtualBox 7.2.18 was installed on the Windows 11 host, and a VM named `Ubuntu-Cloud-Lab` (4 vCPU, 6 GB RAM, 25 GB VDI, NAT networking) was created from the Ubuntu 26.04.1 LTS Desktop ISO image. The guest OS was installed, booted and verified to be running with working network connectivity, and the build tools (`build-essential`) and kernel headers (`linux-headers-7.0.0-31-generic`) were installed successfully.

9. Conclusion

This experiment demonstrated how a Type 2 hypervisor abstracts physical hardware to run an isolated guest operating system from an image file, which is the same principle cloud platforms use to provision virtual machines on demand. The configured Ubuntu VM, with compilers and kernel headers installed, is ready to be used for the following Cloud Computing lab experiments.